End-Term Examination

(CBCS) (SUBJECTIVE TYPE)

B. Tech CSE/IT/CSE-AI, 4th Semester

(May, 2022)

Q.P Code 220539

Subject Code: BIT 202	**Subject: Operating Systems**

Time: 3 Hours **Maximum Marks:** 60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1. **(4*5=20)**

(a) How is a thread different from a process? What do you understand by a multi-threaded process and how is it different from a single-threaded process, show with the help of a diagram?

(b) Explain the concept of deadlock and what are the different conditions that must hold simultaneously for deadlock to occur? How can deadlock be prevented?

(c) For the following set of processes, find the average waiting time & average turnaround time using GANTT Chart using priority scheduling algorithm.

Process	Arrival Time (in secs)	Burst Time (in secs)	Priority
A	0	2	1(highest)
B	1	4	3(3rd highest)
C	2	4	2(2nd highest)
D	3	2	4(lowest)

(d) Assume there are 4 frames and the page reference string is 7,4,1,3,0,1,7,3,4,0,1,3,4,0,3,0,3,3,4,7,0,3. Find the total number of page faults that occur using Least Recently Used(LRU) page replacement approach.

Unit-I

Q2. For the following set of processes, find the average waiting time & average turnaround time using GANTT Chart such that the shortest job is executed first completely. Show how the average waiting time and average turnaround time will change if these processes are executed using shortest remaining job first. **(10)**

Process	Arrival Time (in secs)	Burst Time(in secs)
A	0	2
B	2	9
C	4	2
D	6	3
E	10	1

Q3. For the following set of processes, find the average waiting time & average turnaround time using GANTT Chart. The jobs will execute in round robin fashion and the time slice is given as 3 msec and each context switch takes 1 msec. How many total context switches will be there? **(10)**

Process	Arrival Time (in msecs)	Burst Time(in msecs)
P1	0	5
P2	1	6
P3	2	7
P4	3	8

UNIT-2

Q4. Consider the following page reference string **(10)**
7,4,1,3,0,1,7,3,4,0,1,3,5,3,7,0,1,0,3,1,5,4,5,1,0,7,3,7.
Find out the number of page faults if there are 4 frames, using the optimal replacement algorithm. How will the number of page faults vary in case LRU is used? Explain briefly about the different page replacement algorithms.

Q5. An operating system uses the banker's algorithm for deadlock avoidance when managing the allocation of three resource types X, Y and Z to three processes P0, P1 and P2. The table given below presents the current system state. Here, the Allocation matrix shows the current number of resources of each type allocated to each process and the Max matrix shows the maximum number of resources of each type required by each process during its execution. **(10)**

	Allocation			Max		
	X	Y	Z	X	Y	Z
P0		0	1	8	4	3
P1		2	0	6	2	0
P2		1	1	3	3	3

There are 3 units of type X, 2 units of type Y and 2 units of type Z still available. The system is currently in safe state.

i) What will be the content of the Need matrix?

ii) Is the system in a safe state? If Yes, then what is the safe sequence?

iii) What will happen if process P1 requests one additional instance of resource type X and two instances of resource type Y?

UNIT-3

Q6. Which are the two atomic operations used by semaphores to solve the critical section problem? What are the two main types of semaphores and show how they are implemented with the help of an example? **(10)**

Q7. Which are the classical problems of synchronization? Explain any two in detail with code snippets. **(10)**

UNIT-4

Q8. Describe various file access methods. Also explain how free space is managed. **(10)**

Q9. What is disk scheduling? Explain FCFS and SCAN disk scheduling algorithms. **(10)**